Supplementary Material: A Multi-Axis Empirical Audit of Machine-Learning Intrusion Detection

AUTHOR_CONFIRMATION_REQUIRED[a]

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Use and evidence boundary

This supplement contains reviewer-facing frozen evidence and provenance records. It does not redistribute restricted datasets, raw traffic, private information, closed scientific targets, unauthorized model checkpoints, or the complete research repository. Values below are transcribed from approved artifacts without recomputation. Each quantitative exhibit names its population, metric, condition, qualification, and source hash in `source-manifest.csv`. A source listed as unavailable or restricted remains a dependency; no missing value is inferred.

S1. Reference evaluation, uncertainty, and explanation reliability

The processed CSE-CIC-IDS2018 reference is a rebalanced within-table condition. Its membership counts, threshold roles, operating metrics, calibration records, paired conditional intervals, fixed-recipe seed summaries, LIME reliability diagnostics, and TreeSHAP comparisons remain conditional on the frozen holdout and protocol. No learner-winner claim is introduced. Complete raw arrays and closed targets are not redistributed.

S1.1 Approved reference evidence

Population: processed, rebalanced CSE-CIC-IDS2018 holdout. Metrics: F1, F2, FPR, ROC-AUC, PR-AUC, Brier score, and conditional intervals. Qualification: this condition does not establish natural prevalence, temporal validity, transfer, or deployment utility. The approved source summary is `source/RESULTS_SUMMARY.md`; source hash appears in the manifest. LIME and TreeSHAP reliability evidence is source-traceable in the repository but

Table S1. Reference evidence classification.

Evidence	Frozen interpretation	Qualification
Reference holdout	Processed, rebalanced CSE-CIC-IDS2018; fixed train/validation/holdout roles	Within-table reference only
Operating points	Standard, balanced, and security roles retained separately	Threshold behavior is condition-specific
Calibration and uncertainty	Paired conditional intervals and calibration diagnostics	Conditional on frozen sample, arrays, and implementation
Explanation reliability	LIME local reconstruction/agreement and TreeSHAP comparisons	Stable explanations do not imply fidelity or causality

not reproduced as a new aggregate because the approved panel contains case-level and method-specific records.

Traceability: Stage 29 evidence IDs E29-006–E29-011, E29-015; source artifact registry and approved reference records. Redistribution: summary-level exhibit permitted; raw cases/arrays excluded.

S2. Shortcut, representation, and architecture diagnostics

The seven registered feature subsets are reproduced below from the approved Stage 23 table. Random-natural and chronological-natural populations remain separate. Metrics are native ranking and operating-point metrics at threshold 0.50. Subset effects are sensitivity evidence, not proof of leakage or causality. Placebo controls are shown separately to preserve their distinct schema and interpretation.

Table S2. Seven registered feature-subset results.

Subset	Features	R PR-AUC	C PR-AUC	R ROC-AUC	C ROC-AUC	R F1	C F1	C FPR
FULL	70	.995590	.106215	.998625	.514918	.983867	.000064	.000058
NO_DST_PORT	69	.995260	.102474	.998473	.488864	.983674	.000225	.000135
NO_PORTS	68	.995267	.102473	.998476	.488864	.983684	.000225	.000135
NO_INIT_FWD_WIN_BYTS	69	.993417	.106032	.998192	.509212	.974099	.002546	.000960
NO_FWD_SEG_SIZE_MIN	69	.995576	.105455	.998609	.511682	.983860	.000128	.000081
NO_SUSPICIOUS_GROUP	67	.989601	.104667	.997409	.489393	.963007	.019143	.006496
BEHAVIOR_ONLY	63	.981588	.095540	.996142	.476482	.946547	.029455	.045740

R and C denote random-natural and chronological-natural. Source: `stage23_7_final_synthesis/tables/table_23_1_primary_subset_performance.csv`; source population and hash are in the manifest.

The placebo source population is 2,882,481 random-natural validation rows and 593,780 chronological-natural validation rows, with the source-specific at-

Table S3. Matched placebo controls and interaction intervals.

Placebo	R PR-AUC	C PR-AUC	PR interaction	95% interval	R ROC-AUC	C ROC-AUC
COUNTS	.995587	.106567	.000355	[-.000192,.000571]	.998617	.516785
VOLUME_DIRECTION	.995581	.105759	-.000447	[-.000803,-.000167]	.998617	.513851
IAT	.995580	.105298	-.000908	[-.001854,-.000826]	.998618	.512494
PACKET_SIZE	.991702	.106017	.003690	[.002138,.004160]	.996802	.515720
ACTIVITY	.995584	.106555	.000345	[-.000225,.000601]	.998617	.516519

tack supports retained. Source: `table_23_s1_placebo_subset_performance.csv` and `table_23_s2_placebo_interactions.csv`. Placebo interactions can vary by split; they do not identify one causal shortcut mechanism. Stage 18 graph/attention and Stage 21 CNN/ViT materials remain descriptive or supplement-only under E29-017–E29-019 and are not promoted here.

S3. Cross-dataset contracts and sensitivity

Bridge directions remain separate. The values below are the actual approved direction-specific transfer results; no pooled score is formed. The reverse target is the IDS2018 February 28 population. Bridge62 excludes all eight aggregate flag-count fields, so its published and corrected rows are bitwise identical. Bridge70 retains published and corrected serialization as separate conditions.

Table S4. Direction-specific bridge62/bridge70 transfer metrics.

Direction	Bridge	Semantics	Target prev.	PR-AUC	PR excess	ROC-AUC	Brier
IDS18 → CICIDS17	62	Published/corrected	.196996	.667483	.470486	.733946	.145756
IDS18 → CICIDS17	70	Published	.196996	.663981	.466985	.741891	.142912
IDS18 → CICIDS17	70	Corrected	.196996	.656252	.459256	.744083	.147998
CICIDS17 → IDS18	62	Corrected source	.104847	.108176	.003329	.525167	.105758
CICIDS17 → IDS18	70	Corrected source	.104847	.107419	.002572	.525302	.105652

Table S5. Approved paired bootstrap contrasts.

Direction / comparison	Metric	Difference	95% interval	Excludes zero
Forward: B70 published–B62 published	PR-AUC	-.003502	[-.003856,-.003152]	yes
Forward: B70 published–B62 published	ROC-AUC	.007945	[.007793,.008095]	yes
Forward: B70 corrected–B62 corrected	PR-AUC	-.011230	[-.011619,-.010858]	yes
Forward: B70 corrected–B62 corrected	ROC-AUC	.010137	[.009981,.010303]	yes
Forward: B70 corrected–B70 published	PR-AUC	-.007729	[-.007921,-.007534]	yes
Forward: B70 corrected–B70 published	ROC-AUC	.002192	[.002139,.002246]	yes
Reverse: B70–B62	PR-AUC	-.000757	[-.001058,-.000435]	yes
Reverse: B70–B62	ROC-AUC	.000135	[-.000900,.001263]	no

Each contrast uses 2,000 paired bootstrap replicates, seed 42, conditional on the frozen target and bridge protocol. Sources: the approved bridge62 and bridge70 source tables. The two planned `GROUNDED_S4` cells were cancelled before target opening because exact durable membership was unavailable; no result exists (E29-027).

S4. Prevalence, workload, capacity, and cost

The following approved table is an analytic prior-shift projection. TPR and FPR are held fixed while projected prevalence is set to 0.001. Traffic volume, analyst time, and relative cost are scenarios, not field observations.

Table S6. Frozen 0.1% prevalence operational translation.

Cell	Precision	TPR	FPR	PPV	FP/day	Hours/day
Stage22 Random	.999775	.968458	.0000346	.965572	34.566	33.466
Stage22 Chronological	.060606	.0000321	.0000583	.000551	58.323	1.945
IDS18 to CICIDS17 B62	.927118	.212924	.004106	.049344	4106.292	143.981
IDS18 to CICIDS17 B70 published	.940228	.227838	.003553	.060313	3553.302	126.046
IDS18 to CICIDS17 B70 corrected	.909155	.145044	.003556	.039233	3555.502	123.356
CICIDS17 to IDS18 B62	.029268	.000386	.001498	.000258	1497.581	49.932
CICIDS17 to IDS18 B70	.032609	.000386	.001340	.000288	1339.544	44.664

The full source additionally records true-alert volume, total alerts, cost decisions, required-FPR curves, capacity tiers, and break-even conditions. Those tables remain source-traceable but are not collapsed into a new summary. Sources: `table25_1_standard_operational_translation_0p1pct.csv`, Stage 25 publication tables; E29-028–E29-031. No field-performance claim follows.

S5. Deployment profiling

Measurements cover prepared model input through materialized probabilities on recorded hardware/software. They exclude capture, flow extraction, external representation construction, alert aggregation, and analyst response. Unsupported, timeout, and resource-limited paths remain statuses.

Source: `T26_CPU_WARM_INFERENCE.csv`; Stage 26 hardware/software receipt and publication tables. Confidence intervals, throughput, memory, GPU matches, Pareto records, and backend statuses remain in the source artifact. These are hardware-specific component measurements, not end-to-end deployment latency (E29-032–E29-033).

Table S7. Representative approved CPU warm-inference measurements.

Target	CPU	Batch	Status	n	p50 ms	p95 ms
XGBoost tuned	1 core	1	PASS	200	.4397	.5937
XGBoost tuned	1 core	256	PASS	100	2.4264	2.7338
LightGBM tuned	1 core	1	PASS	200	.5368	.6488
CatBoost tuned	1 core	1	PASS	200	.2054	.2392
5-checkpoint voting	1 core	1	PASS	200	8.8806	11.3027
5-checkpoint voting	1 core	8192	TIMEOUT_RESOURCE_LIMIT	0	–	–
Packet-image CNN	1 core	256	PASS	100	4580.8021	4987.9484

S6. LOAO eligibility, family results, and stability controls

LOAO requires zero training/validation exposure to the target family, day-atomic ordering, and sufficient target support. DDoS, Port Scan, Web Attack, Bot, and Infiltration were eligible; Infiltration remains descriptive at 36 positives. DOS and AUTH_BRUTE_FORCE were structurally ineligible.

Table S8. Family-level LOAO primary metrics.

Family	Learner	Support	Rows	ROC-AUC	95% CI	PR-AUC	Claim status
Bot	XGBoost	1966	416288	.3224	[.3102,.3354]	.0033	eligible
Bot	LightGBM	1966	416288	.5591	[.5521,.5660]	.0049	eligible
DDoS	XGBoost	128027	542349	.9982	[.9981,.9983]	.9925	eligible
DDoS	LightGBM	128027	542349	.9986	[.9985,.9987]	.9940	eligible
Infiltration	XGBoost	36	456788	.7816	[.7473,.8142]	.000181	descriptive only
Infiltration	LightGBM	36	456788	.7537	[.7239,.7865]	.000156	descriptive only
Port Scan	XGBoost	158930	573252	.5506	[.5487,.5524]	.3622	eligible
Port Scan	LightGBM	158930	573252	.7559	[.7546,.7570]	.4191	eligible
Web Attack	XGBoost	2180	458932	.9693	[.9660,.9726]	.7206	eligible
Web Attack	LightGBM	2180	458932	.9901	[.9887,.9914]	.7605	eligible

Source: `stage27_final_primary_metrics.csv`; confidence intervals are family-specific conditional intervals. No pooled future-attack score is formed.

Table S9. Five-seed Stage 22 stability summary.

Unit	Metric	Mean	Median	Minimum	Maximum
Random-natural	ROC-AUC	.517615	.519090	.506288	.524974
Random-natural	PR-AUC	.259921	.261240	.254641	.262814
Chronological-natural	ROC-AUC	.820900	.819597	.808632	.831170
Chronological-natural	PR-AUC	.638752	.644872	.587465	.676863

Source: `stage28_final_stage22_shared_holdout_five_seed_summary.csv` and `stage28_final_loao_stability_registry.csv`. Five-of-five ordering is descriptive, not a population guarantee; seed variation is not combined with bootstrap uncertainty.

Table S10. Evidence classification and reproduction boundary.

Class	Examples	Boundary
Rerunnable from frozen public/source artifacts	Static parsing, source hashes, LaTeX build, registry checks	Does not recreate restricted data or closed targets
Static-only / reviewer-facing	Approved tables, claims, figures, and manifests	Values copied; no scientific recomputation
Archival / environment-bound	Historical model receipts, hardware timings, immutable execution records	Requires recorded environment and artifacts
Unavailable or restricted	Raw traffic, private information, unauthorized checkpoints, closed targets, missing historical environments	Not redistributed; no replacement value
Cancelled / unsupported	`GROUNDED_S4` cells, ineligible families, unsupported backends	No result exists or status is not a measurement

S7. Provenance, equivalence, and reproduction

Traceability: Stage 29 evidence IDs E29-020–E29-041, `source_artifact_registry.csv`, `claim_to_artifact_map.csv`, and the manifest hashes. Some environments remain `VERSION_NOT_PROVEN`; the package supports claim-level traceability, not complete end-to-end regeneration of every result. Redistribution is limited to approved documentation, tables, and public figure assets.

Main/supplement cross-reference audit

S1 supports the main reference condition and explanation-reliability qualifications. S2 supports the shortcut sensitivity discussion. S3 supports the directional bridge discussion and Table 3. S4 supports the prior-shift operational discussion and Table 4. S5 supports deployment-boundary claims and Figure 5. S6 supports LOAO qualifications and stability claims without duplicating the main Figure 6. S7 supports the provenance and limitation statements. Supplement-only or unavailable evidence is not promoted into the main conclusions.

Supplement production QA

Ten quantitative tables are included; no supplementary figures are included, avoiding duplication of the main Figure 6 panels. All exhibits preserve

source populations, metric definitions, thresholds or ranking conditions, uncertainty semantics, support, and qualifications. The source manifest records SHA-256 values. Main and supplement PDFs compile independently. Final author metadata, redistribution review, and venue portal confirmation remain required.